

SATURN'S MOONS

The moons of Saturn are divided into three groups. The first consists of the major moons, which are large and spherical. The second group, the inner moons, are smaller and irregularly shaped. Members of both these groups orbit within or outside the ring system. The third set of moons lies far beyond the other two—the most distant orbit over 15 million miles (25 million km) from Saturn. These irregularly shaped moons are tiny, just a few miles to tens of miles across. They have inclined orbits, which suggests that they are captured objects. From Earth, Saturn's moons appear as little more than disks of light, but Voyager and Cassini revealed many of them as worlds in their own right.

DWARFED BY SATURN

Saturn's moons, such as Tethys (top) and Dione (below), are not only small compared to their host planet but, with the exception of Titan, they are all smaller than Earth's Moon.

INNER MOON

Prometheus

DISTANCE FROM SATURN 86,539 miles (139,353 km)

ORBITAL PERIOD 0.61 Earth days

LENGTH 84 miles (136 km)

Prometheus is a small, elongated moon orbiting just inside the multi-stranded F ring. Along with Pandora, it is a "shepherd" of the F ring.

Cassini images of Prometheus and the F ring show them to be linked by a fine thread of material, produced as Prometheus pulls particles out of the ring. The moon's long axis points toward Saturn.



SHEPHERD MOON

INNER MOON

Janus

DISTANCE FROM SATURN 94,120 miles (151,472 km)

ORBITAL PERIOD 0.69 Earth days

LENGTH 126 miles (203 km)

Heavily cratered and irregularly shaped, Janus orbits Saturn just beyond the F ring and only 30 miles (50 km) farther away than its co-orbital moon, Epimetheus. Its existence was first reported in December 1966, and it was named after the Roman god Janus, who could look forward and back at the same time. Yet it was only confirmed as a moon in February 1980, after Voyager 1 data had been studied.



BEYOND THE F RING

INNER MOON

Epimetheus

DISTANCE FROM SATURN 94,089 miles (151,422 km)

ORBITAL PERIOD 0.69 Earth days

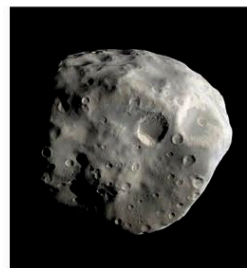
LENGTH 81 miles (130 km)

Occasionally, moons orbit a planet within about 30 miles (50 km) of each other. They are described as co-orbital since they virtually share an orbit. The two moons Epimetheus and Janus

CO-ORBITAL MOON

Epimetheus orbits against the backdrop of Saturn's rings, which are seen nearly edge-on in this view taken by Cassini's narrow-angle camera on February 18, 2005.

(below left), which orbit just beyond the F ring, are such a pair. They swap orbits every 4 years, taking turns being slightly closer to Saturn. Epimetheus is a lumpy moon just 17 miles (28 km) longer than it is wide or deep, and it is one of 16 moons that lie within the ring system. Epimetheus is in synchronous rotation—that is, it keeps the same face toward Saturn at all times because its rotation and orbital periods are the same. As it orbits Saturn, it works as a shepherd moon, confining the ring particles within the F ring. Prometheus (left) works

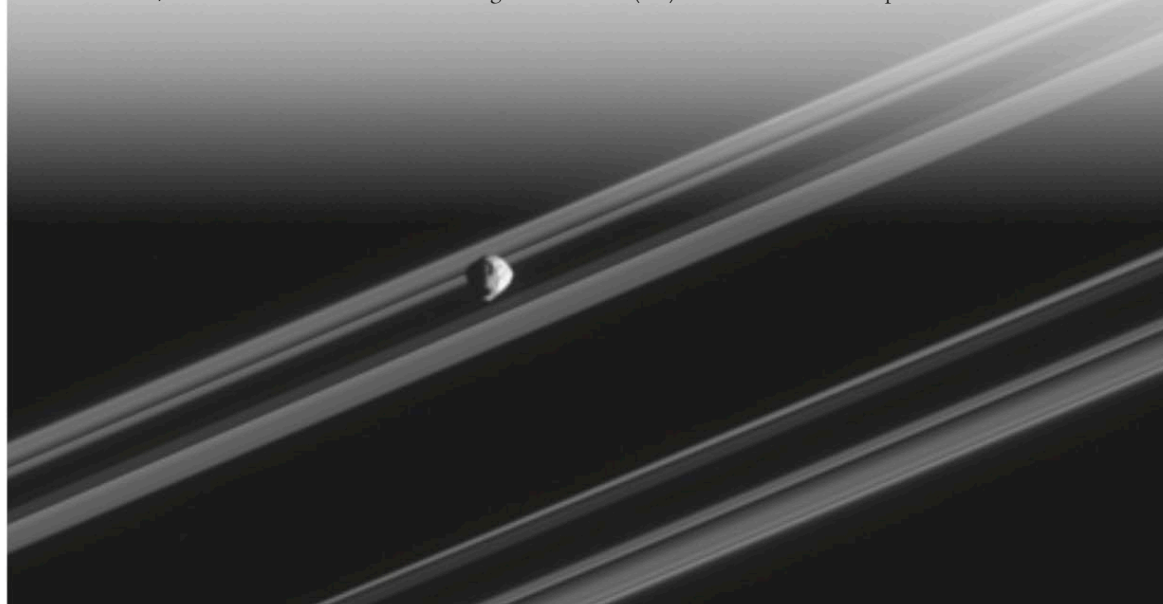


BATTERED SURFACE

Epimetheus (left) and its co-orbital moon, Janus, are believed to be the remnants of a larger object that was broken apart by an impact.

the same way on the inner side of the ring. The existence of Epimetheus was

suspected in 1967, but was not confirmed until February 26, 1980. It was one of eight moons discovered in Voyager data that year. The moon is named after a Titan, the family of giants in Greek mythology who once ruled the Earth. Prometheus was one of Epimetheus's five brothers.



TINY MOON

Methone

DISTANCE FROM SATURN 120,550 miles (194,000 km)

ORBITAL PERIOD 1 Earth day

DIAMETER 1.65 miles (3 km)

Two small moons orbiting between the major moons Mimas and Enceladus were discovered in 2004 in data collected by the Cassini probe. As with all such discoveries, the moons were initially identified by numerical designations (S/2004 S1 and S/2004 S2). The two moons are now known as Methone and

Pallene. They were not discovered by chance, but were identified in images taken as part of a search for new moons within this region around Saturn. The contrast of the images was enhanced to increase visibility. Methone was visited by the Cassini spacecraft in 2012, which gave scientists a better look at the egg-shaped moon.

SMOOTH SATELLITE

In 2012, the Cassini spacecraft made two approaches to Methone, getting as close as 1,200 miles (1,900 km) to the moon. Cassini images show Methone's surface to be relatively smooth and free from visible craters.

